DBMS LAB ASSIGNMENT 4, 5 and 6

Github link : https://github.com/Wruthambara/DBMS_L1_L2.git

```sql
CREATE TABLE Students (

    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(20),
    Balance DECIMAL(10,2),
    Admission_Date DATE,
    Marks INT,
    Fee_Amount DECIMAL(10,2),
    Date_of_Birth DATE
);

INSERT INTO students (Student_ID, Student_Name, Class, Balance, Admission_Date, Marks, Fee_Amount, Date_of_Birth)
VALUES
(301, 'Ruthu Gowda', '10A', 68000, '2023-06-08', 97, 55000, '2008-03-15'),
(302, 'Mrudula Bhat', '10B', 92000, '2023-06-12' ,90, 60000, '2008-07-22'),
(303, 'Dhrithi Sharam', '10A', 47000, '2023-06-15', 87, 48000, '2009-01-10'),
(304, 'Vashanvi Guptha', '10C', 85000, '2023-06-18', 96, 52000, '2008-10-05'),
(305, 'simtha Malhotra', '10B', 59000, '2023-06-20', 91, 45000, '2009-02-28');

SELECT * FROM students;



SELECT Student_Name,
UPPER(Student_Name) AS Uppercase_Name
FROM Students;

SELECT Student_Name,
LOWER(Student_Name) AS Lowercase_Name
FROM Students;

SELECT Student_Name,
LENGTH(Student_Name) AS Name_Length
FROM Students;

SELECT CONCAT(Student_Name,' - Class ', Class) AS Student_Details
FROM Students;

SELECT Student_Name,
SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM Students;
```

```sql
SELECT Student_Name,
Marks,
ROUND(Marks * 1.05) AS Bonus_Marks
FROM Students;


SELECT Student_Name,
Marks / 10.0 AS Marks_Value,


CEIL(Marks / 10.0) AS Rounded_Up
FROM Students;


SELECT Student_Name,
Marks / 10.0 AS Marks_Value,
FLOOR(Marks / 10.0) AS Rounded_Down

FROM Students;


SELECT Student_Name,
Marks,
MOD(Marks, 10) AS Remainder
FROM Students;

SELECT Student_Name,
Marks,
ABS(Marks - 80) AS Difference_From_80
FROM Students;




SELECT Student_Name,
Date_of_Birth,
YEAR(Date_of_Birth) AS Birth_Year
FROM Students;


SELECT Student_Name,
Date_of_Birth,
MONTH(Date_of_Birth) AS Birth_Month
FROM Students;
```

```sql
SELECT Student_Name,
Date_of_Birth,
MONTH(Date_of_Birth) AS Birth_Month
FROM Students;


SELECT Student_Name,
Date_of_Birth,
DAY(Date_of_Birth) AS Birth_Day
FROM Students;


SELECT Student_Name,
Date_of_Birth,
TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM Students;


SELECT CURDATE() AS Today;



SELECT COUNT(*) AS Total_Students
FROM Students;


SELECT SUM(Marks) AS Total_Marks
FROM Students;


SELECT AVG(Marks) AS Average_Marks
FROM Students;


SELECT MAX(Marks) AS Highest_Marks
FROM Students;


SELECT MIN(Marks) AS Lowest_Marks
FROM Students;
```

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_Name	Uppercase_Name
Ruthu Gowda	RUTHU GOWDA
Mrudula Bhat	MRUDULA BHAT
Dhrithi Sharam	DHRITHI SHARAM
Vashanvi Guptha	VASHANVI GUPTHA
simtha Malhotra	SIMTHA MALHOTRA

Student_Name	Lowercase_Name
Ruthu Gowda	ruthu gowda
Mrudula Bhat	mrudula bhat
Dhrithi Sharam	dhrithi sharam
Vashanvi Guptha	vashanvi guptha
simtha Malhotra	simtha malhotra

Student_Name	Name_Length
Ruthu Gowda	11
Mrudula Bhat	12
Dhrithi Sharam	14
Vashanvi Guptha	15
simtha Malhotra	15

Student_Details
Ruthu Gowda - Class 10A
Mrudula Bhat - Class 10B
Dhrithi Sharam - Class 10A
Vashanvi Guptha - Class 10C
simtha Malhotra - Class 10B

Student_Name	First_Five_Characters
Ruthu Gowda	Ruthu
Mrudula Bhat	Mrudu
Dhrithi Sharam	Dhrit
Vashanvi Guptha	Vasha
simtha Malhotra	simth

Student_Name	Marks	Bonus_Marks
Ruthu Gowda	97	102
Mrudula Bhat	90	95
Dhrithi Sharam	87	91
Vashanvi Guptha	96	101
simtha Malhotra	91	96

Student_Name	Marks_Value	Rounded_Up
Ruthu Gowda	9.7000	10
Mrudula Bhat	9.0000	9
Dhrithi Sharam	8.7000	9
Vashanvi Guptha	9.6000	10
simtha Malhotra	9.1000	10

Student_Name	Marks_Value	Rounded_Down
Ruthu Gowda	9.7000	9
Mrudula Bhat	9.0000	9
Dhrithi Sharam	8.7000	8
Vashanvi Guptha	9.6000	9
simtha Malhotra	9.1000	9

Student_Name	Marks	Remainder
Ruthu Gowda	97	7
Mrudula Bhat	90	0
Dhrithi Sharam	87	7
Vashanvi Guptha	96	6
simtha Malhotra	91	1

Student_Name	Marks	Difference_From_80
Ruthu Gowda	97	17
Mrudula Bhat	90	10
Dhrithi Sharam	87	7
Vashanvi Guptha	96	16
simtha Malhotra	91	11

Student_Name	Date_of_Birth	Birth_Year
Ruthu Gowda	2008-03-15	2008
Mrudula Bhat	2008-07-22	2008
Dhrithi Sharam	2009-01-10	2009
Vashanvi Guptha	2008-10-05	2008
simtha Malhotra	2009-02-28	2009

Student_Name	Date_of_Birth	Birth_Month
Ruthu Gowda	2008-03-15	3
Mrudula Bhat	2008-07-22	7
Dhrithi Sharam	2009-01-10	1
Vashanvi Guptha	2008-10-05	10
simtha Malhotra	2009-02-28	2

Student_Name	Date_of_Birth	Birth_Day
Ruthu Gowda	2008-03-15	15
Mrudula Bhat	2008-07-22	22
Dhrithi Sharam	2009-01-10	10
Vashanvi Guptha	2008-10-05	5
simtha Malhotra	2009-02-28	28

Student_Name	Date_of_Birth	Age
Ruthu Gowda	2008-03-15	18
Mrudula Bhat	2008-07-22	18
Dhrithi Sharam	2009-01-10	17
Vashanvi Guptha	2008-10-05	17
simtha Malhotra	2009-02-28	17

Today
2026-10-02

Total_Students
5

Total_Marks
461

Average_Marks
92.2000

Highest_Marks
97

Lowest_Marks
87

```sql
SELECT Student_Name,
       CAST(Marks AS DECIMAL(10,2)) AS Converted_Marks
FROM Students;

SELECT Student_Name,
       CAST(Balance AS CHAR) AS Balance_As_Text
FROM Students;

SELECT Student_Name,
       Fee_Amount + 5000 AS Increased_Fee
FROM Students;

SELECT Student_Name,
       Fee_Amount + 5000 AS Increased_Fee
FROM Students;

SELECT Student_Name,
       Fee_Amount - 5000 AS Reduced_Fee
FROM Students;

SELECT Student_Name,
       Fee_Amount * 2 AS Double_Fee
FROM Students;

SELECT Student_Name,
       Fee_Amount / 2 AS Half_Fee
FROM Students;



SELECT *
FROM Students
WHERE Marks > 90;

SELECT *
FROM Students
WHERE Marks < 90;

SELECT *
FROM Students
WHERE Class = '10A';

SELECT *
FROM Students
```

STDIN

Input for the program (Optional)

Output:

```
+-----------------+-----------------+
| Student_Name    | Converted_Marks |
+-----------------+-----------------+
| Ruthu Gowda     |           97.00 |
| Mrudula Bhat    |           90.00 |
| Dhrithi Sharam  |           87.00 |
| Vashanvi Guptha |           96.00 |
| simtha Malhotra |           91.00 |
+-----------------+-----------------+
+-----------------+-----------------+
| Student_Name    | Balance_As_Text |
+-----------------+-----------------+
| Ruthu Gowda     | 68000.00        |
| Mrudula Bhat    | 92000.00        |
| Dhrithi Sharam  | 47000.00        |
| Vashanvi Guptha | 85000.00        |
| simtha Malhotra | 59000.00        |
+-----------------+-----------------+
+-----------------+---------------+
| Student_Name    | Increased_Fee |
+-----------------+---------------+
| Ruthu Gowda     |      60000.00 |
| Mrudula Bhat    |      65000.00 |
| Dhrithi Sharam  |      53000.00 |
| Vashanvi Guptha |      57000.00 |
| simtha Malhotra |      50000.00 |
+-----------------+---------------+
+-----------------+---------------+
| Student_Name    | Increased_Fee |
+-----------------+---------------+
| Ruthu Gowda     |      60000.00 |
```

```sql
SELECT *
FROM Students
WHERE Marks <> 90;

SELECT *
FROM Students
WHERE Marks >= 90;

SELECT *
FROM Students
WHERE Marks <= 90;

SELECT *
FROM Students
WHERE Marks > 90 AND Balance > 60000;

SELECT *
FROM Students
WHERE Marks > 95 OR Class = '10B';

SELECT *
FROM Students
WHERE NOT Class = '10A';

SELECT *
FROM Students
WHERE NOT Class = '10A';


SELECT *
FROM Students
WHERE Marks BETWEEN 90 AND 97;

SELECT *
FROM Students
WHERE Class IN ('10A', '10B');

SELECT *
FROM Students
WHERE Class NOT IN ('10A', '10B');
```

STDIN

Input for the program (Optional)

Output:

```
+------------------+----------------+
| Student_Name     | Increased_Fee  |
+------------------+----------------+
| Ruthu Gowda      |       60000.00 |
| Mrudula Bhat     |       65000.00 |
| Dhrithi Sharam   |       53000.00 |
| Vashanvi Guptha  |       57000.00 |
| simtha Malhotra  |       50000.00 |
+------------------+----------------+
+------------------+--------------+
| Student_Name     | Reduced_Fee  |
+------------------+--------------+
| Ruthu Gowda      |     50000.00 |
| Mrudula Bhat     |     55000.00 |
| Dhrithi Sharam   |     43000.00 |
| Vashanvi Guptha  |     47000.00 |
| simtha Malhotra  |     40000.00 |
+------------------+--------------+
+------------------+-------------+
| Student_Name     | Double_Fee  |
+------------------+-------------+
| Ruthu Gowda      |   110000.00 |
| Mrudula Bhat     |   120000.00 |
| Dhrithi Sharam   |    96000.00 |
| Vashanvi Guptha  |   104000.00 |
| simtha Malhotra  |    90000.00 |
+------------------+-------------+
```

```sql
FROM Students
WHERE Class IN ('10A', '10B');

SELECT *
FROM Students
WHERE Class NOT IN ('10A', '10B');

SELECT *
FROM Students
WHERE Student_Name LIKE 'R%';

SELECT *
FROM Students
WHERE Student_Name LIKE '%a';

SELECT *
FROM Students
WHERE Balance IS NULL;

SELECT *
FROM Students
WHERE Balance IS NOT NULL;



SELECT Student_Name, Class
FROM Students
WHERE Class = '10A'

UNION

SELECT Student_Name, Class
FROM Students
WHERE Class = '10B';

SELECT Student_Name, Class
FROM Students
WHERE Class = '10A'

UNION ALL

SELECT Student_Name, Class
FROM Students
WHERE Class = '10B';
```

STDIN

Input for the program (Optional)

Output:

```
+------------------+--------------+
| Student_Name     | Half_Fee     |
+------------------+--------------+
| Ruthu Gowda      | 27500.000000 |
| Mrudula Bhat     | 30000.000000 |
| Dhrithi Sharam   | 24000.000000 |
| Vashanvi Guptha  | 26000.000000 |
| simtha Malhotra  | 22500.000000 |
+------------------+--------------+
+------------+-----------------+-------+----------+----------------+-------+------------+---------------+
| Student_ID | Student_Name    | Class | Balance  | Admission_Date | Marks | Fee_Amount | Date_of_Birth |
+------------+-----------------+-------+----------+----------------+-------+------------+---------------+
|        301 | Ruthu Gowda     | 10A   | 68000.00 | 2023-06-08     |    97 |   55000.00 | 2008-03-15    |
|        304 | Vashanvi Guptha | 10C   | 85000.00 | 2023-06-18     |    96 |   52000.00 | 2008-10-05    |
|        305 | simtha Malhotra | 10B   | 59000.00 | 2023-06-20     |    91 |   45000.00 | 2009-02-28    |
+------------+-----------------+-------+----------+----------------+-------+------------+---------------+
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
| Student_ID | Student_Name   | Class | Balance  | Admission_Date | Marks | Fee_Amount | Date_of_Birth |
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
|        303 | Dhrithi Sharam | 10A   | 47000.00 | 2023-06-15     |    87 |   48000.00 | 2009-01-10    |
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
| Student_ID | Student_Name   | Class | Balance  | Admission_Date | Marks | Fee_Amount | Date_of_Birth |
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
|        301 | Ruthu Gowda    | 10A   | 68000.00 | 2023-06-08     |    97 |   55000.00 | 2008-03-15    |
|        303 | Dhrithi Sharam | 10A   | 47000.00 | 2023-06-15     |    87 |   48000.00 | 2009-01-10    |
+------------+----------------+-------+----------+----------------+-------+------------+---------------+
+------------+-----------------+-------+----------+----------------+-------+------------+---------------+
| Student_ID | Student_Name    | Class | Balance  | Admission_Date | Marks | Fee_Amount | Date_of_Birth |
+------------+-----------------+-------+----------+----------------+-------+------------+---------------+
|        301 | Ruthu Gowda     | 10A   | 68000.00 | 2023-06-08     |    97 |   55000.00 | 2008-03-15    |
```

Output:

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Output:

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15
302	Mrudula Bhat	10B	92000.00	2023-06-12	90	60000.00	2008-07-22
303	Dhrithi Sharam	10A	47000.00	2023-06-15	87	48000.00	2009-01-10
305	simtha Malhotra	10B	59000.00	2023-06-20	91	45000.00	2009-02-28

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
304	Vashanvi Guptha	10C	85000.00	2023-06-18	96	52000.00	2008-10-05

Student_ID	Student_Name	Class	Balance	Admission_Date	Marks	Fee_Amount	Date_of_Birth
301	Ruthu Gowda	10A	68000.00	2023-06-08	97	55000.00	2008-03-15

Output:

```
+------------+-----------------+-------+----------+----------------+-------+------------+---------------
|        301 | Ruthu Gowda     | 10A   | 68000.00 | 2023-06-08     |    97 |   55000.00 | 2008-03-15
|        304 | Vashanvi Guptha | 10C   | 85000.00 | 2023-06-18     |    96 |   52000.00 | 2008-10-05
|        305 | simtha Malhotra | 10B   | 59000.00 | 2023-06-20     |    91 |   45000.00 | 2009-02-28
+------------+-----------------+-------+----------+----------------+-------+------------+---------------
+------------+-----------------+-------+----------+----------------+-------+------------+---------------
| Student_ID | Student_Name    | Class | Balance  | Admission_Date | Marks | Fee_Amount | Date_of_Birth
+------------+-----------------+-------+----------+----------------+-------+------------+---------------
|        301 | Ruthu Gowda     | 10A   | 68000.00 | 2023-06-08     |    97 |   55000.00 | 2008-03-15
|        302 | Mrudula Bhat    | 10B   | 92000.00 | 2023-06-12     |    90 |   60000.00 | 2008-07-22
|        303 | Dhrithi Sharam  | 10A   | 47000.00 | 2023-06-15     |    87 |   48000.00 | 2009-01-10
|        304 | Vashanvi Guptha | 10C   | 85000.00 | 2023-06-18     |    96 |   52000.00 | 2008-10-05
|        305 | simtha Malhotra | 10B   | 59000.00 | 2023-06-20     |    91 |   45000.00 | 2009-02-28
+------------+-----------------+-------+----------+----------------+-------+------------+---------------
+-----------------+-------+
| Student_Name    | Class |
+-----------------+-------+
| Ruthu Gowda     | 10A   |
| Dhrithi Sharam  | 10A   |
| Mrudula Bhat    | 10B   |
| simtha Malhotra | 10B   |
+-----------------+-------+
+-----------------+-------+
| Student_Name    | Class |
+-----------------+-------+
| Ruthu Gowda     | 10A   |
| Dhrithi Sharam  | 10A   |
| Mrudula Bhat    | 10B   |
| simtha Malhotra | 10B   |
+-----------------+-------+
```